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EXPT.NO - 12

TITLE - STACK using ARRAYS - PUSH(),POP() AND DISPLAY() METHODS

```
In [1]: 1 def push(stack, item, max_size):
2         if len(stack) >= max_size:
3             print("STACK OVERFLOW")
4         else:
5             stack.append(item)
6             print("PUSHED", item,"TO STACK")
7
8     def pop(stack):
9         if not stack:
10            print("STACK UNDERFLOW")
11        else:
12            item = stack.pop()
13            print("POPED", item,"FROM STACK")
14            return item
15
16    def display(stack):
17        if not stack:
18            print("STACK IS EMPTY")
19        else:
20            print("STACK ELEMENTS ARE:")
21            for item in reversed(stack):
22                print(item)
23
24
25    #INPUT
26    max_size = int(input("ENTER THE MAXIMUM SIZE OF THE STACK:"))
27    stack = []
28
29    while True:
30        print("=====")
31        print("\n1. Push\n2. Pop\n3. Display\n4. Exitt")
32        choice = int(input("\n ENTER YOUR CHOICE:"))
33
34        if choice == 1:
35            item = int(input("ENTER THE ITEM TO PUSH : "))
36            push(stack,item,max_size)
37        elif choice == 2:
38            pop(stack)
39        elif choice == 3:
40            display(stack)
41        elif choice == 4:
42            break
43        else:
44            print("INVALID CHOICE PLEASE TRY AGAIN.....")
45
```

```
ENTER THE MAXIMUM SIZE OF THE STACK:3
=====

1. Push
2. Pop
3. Display
4. Exitt

ENTER YOUR CHOICE:1
ENTER THE ITEM TO PUSH : 10
PUSHED 10 TO STACK
=====

1. Push
2. Pop
3. Display
4. Exitt

ENTER YOUR CHOICE:1
ENTER THE ITEM TO PUSH : 10
```

In []:

1

